



HITEC UNIVERSITY, TAXILA

DEPARTMENT OF COMPUTER ENGINEERING

Indigenously Developed AI Based Parental Control Router - KidSafeNet

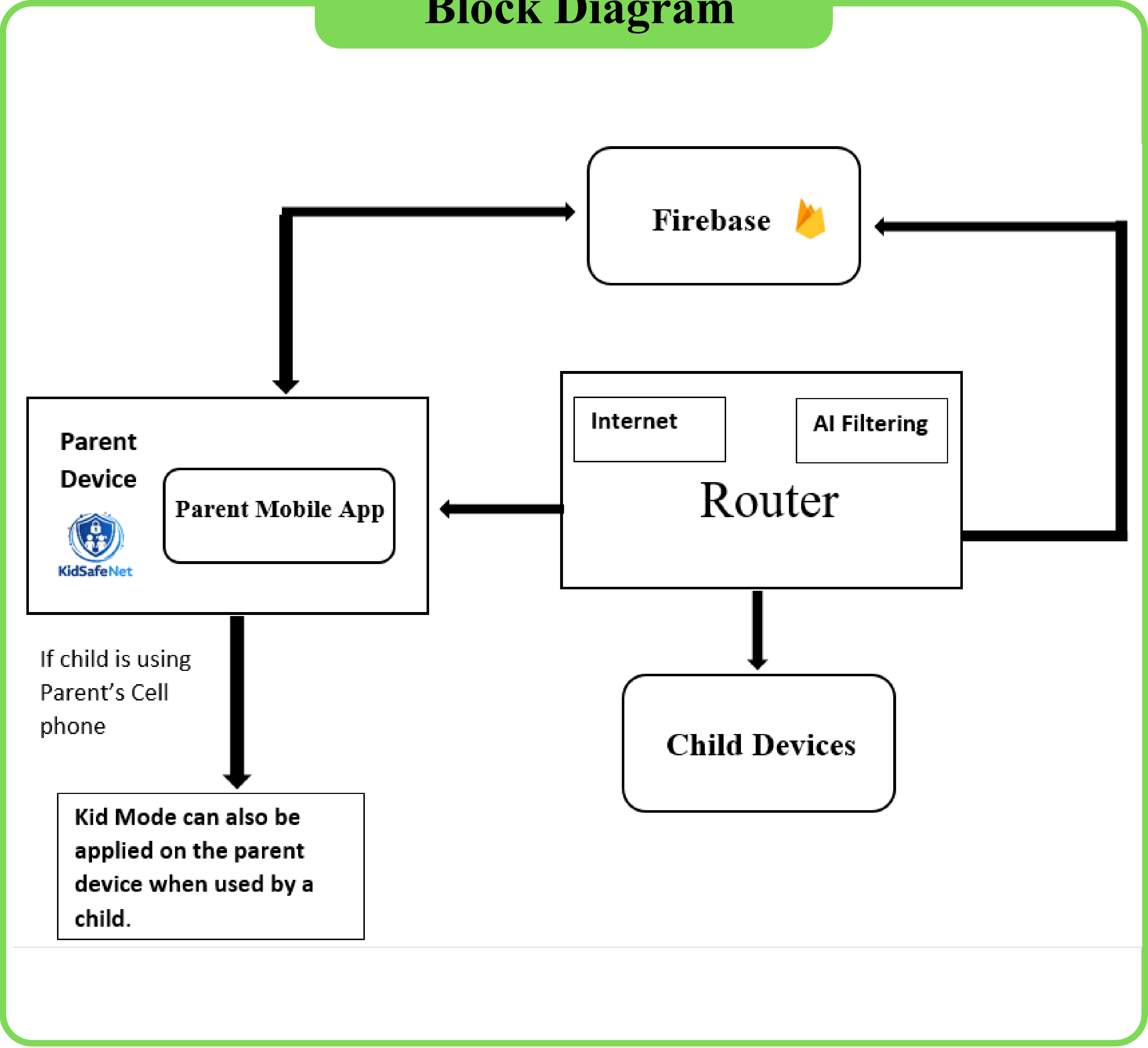
Abstract

KidSafeNet is an indigenously developed AI-based parental control solution designed to enable a safer digital environment for children. The system operates at the network level and monitors internet activity to identify inappropriate, and addictive online content. When Kid Mode is enabled, intelligent filtering mechanisms automatically block inappropriate content and redirect users toward safe and educational resources. When Kid Mode is disabled, normal internet access is restored behave like normal internet provider Router. A dedicated mobile application allows parents to remotely enable or disable Kid Mode and monitor system activity in real time. By providing centralized protection for all connected devices, KidSafeNet offers a practical and effective solution for promoting safer and more responsible internet usage among children.

Methodology

The proposed system monitors internet activity at the network level to provide safer internet usage for children. The system analyzes browsing activity and detects inappropriate or addictive online content such as short videos and inappropriate videos. When inappropriate content is detected, the system automatically blocks or restricts access and redirects users toward safe and educational content. The complete system is implemented using Raspberry Pi 5 with real-time monitoring and filtering features. A mobile application is also developed to help parents monitor activities, manage controls, and promote responsible internet usage for children. Our system operates in two modes: KidMode ON and KidMode OFF. Since not everyone in a household requires internet restrictions, these modes provide flexibility for different users. When KidMode ON is enabled, all monitoring and filtering features are activated. When KidMode OFF is selected, the system functions as a normal internet router and provides unrestricted internet access.

Block Diagram

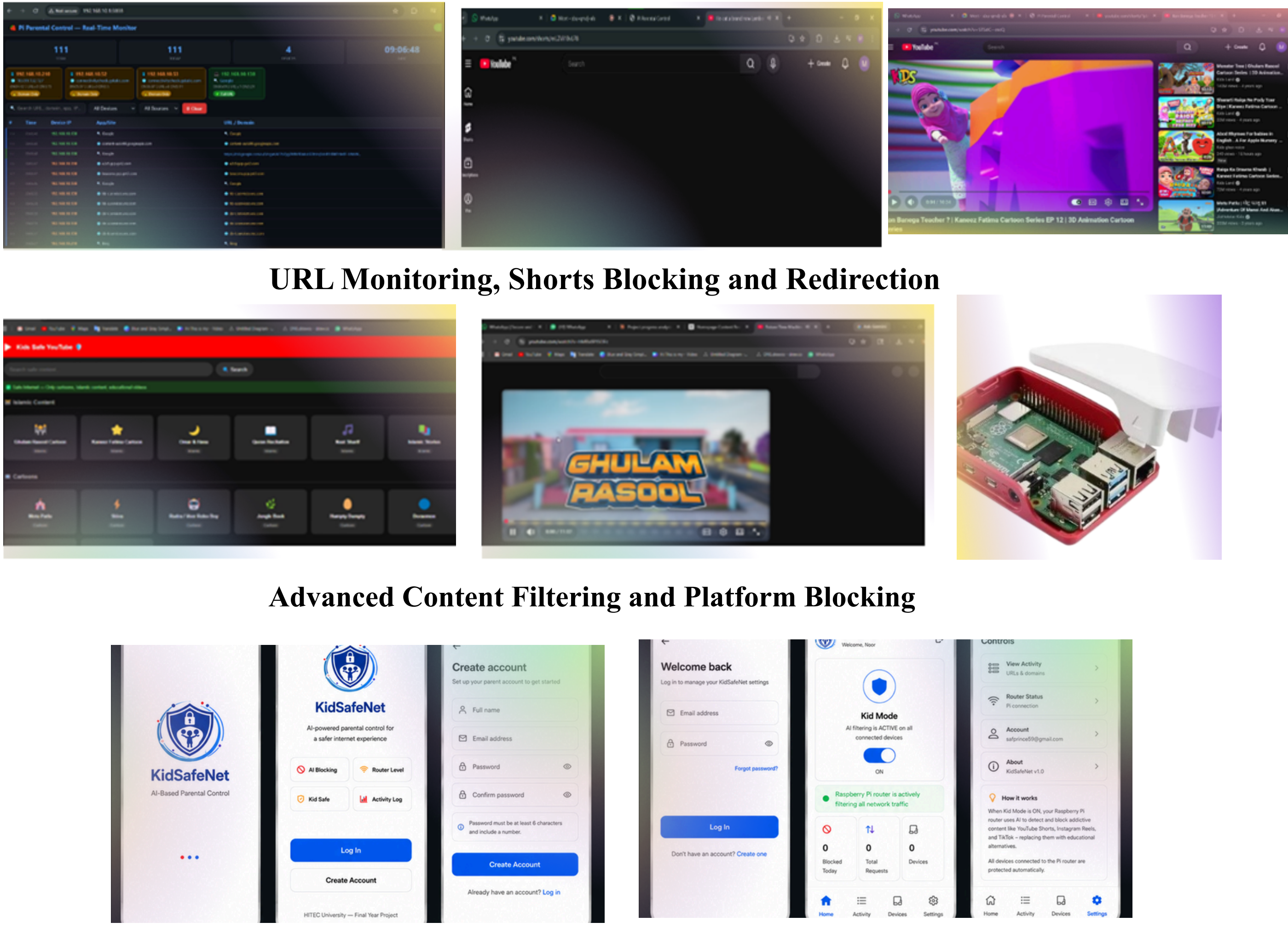


Sustainable Development Goal



Results

- Real-time monitoring of internet traffic through DNS and HTTPS URL analysis.
- AI-based content filtering implemented using CatBoost and RoBERTa models.
- Detection and blocking of inappropriate and addictive online content.
- Blocking of YouTube Shorts and other short-form content at the network level.
- Smart redirection of restricted content toward safe and educational alternatives.
- Centralized parental control for all connected devices.
- Real-time synchronization and activity logging through Firebase integration.
- Development of a Flutter-based mobile application for remote parental monitoring and control.
- Flexible operation through KidMode ON and KidMode OFF functionality.
- Deployment of the complete KidSafeNet system on Raspberry Pi 5 with real-time filtering capabilities.



Conclusion

KidSafeNet provides a practical and intelligent solution for child online safety by combining AI-based content understanding, smart blocking, educational redirection, and a dedicated mobile application. Through its indigenously developed router and app, the system promotes safer, smarter, and more responsible internet usage for children.

Scan Me



Unit Cost:110,000

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